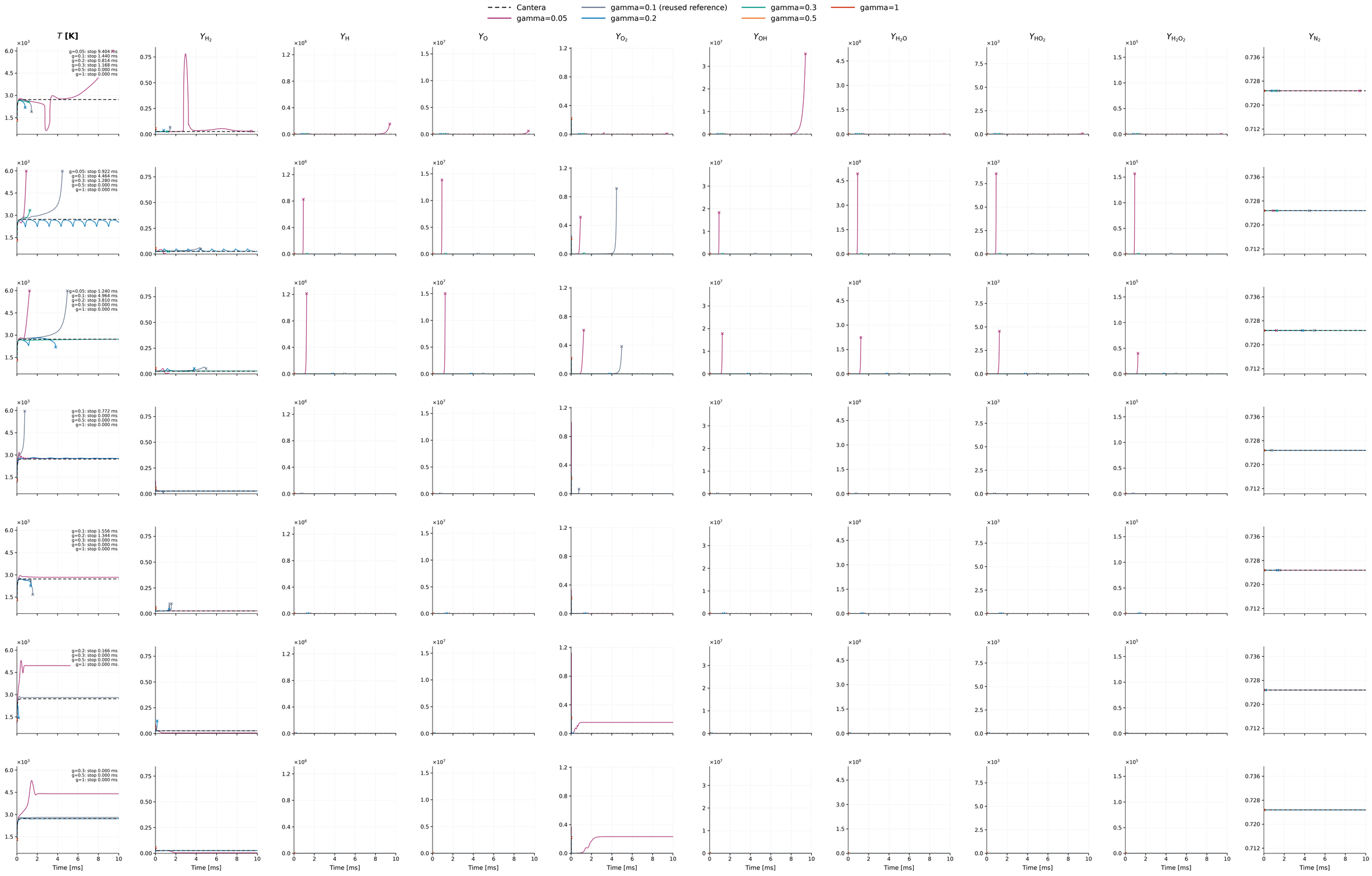
Box-Cox gamma sweep: Cantera vs NN | Test IC 04

T_0=1317.5 K P_0=1.244 atm ϕ_0=1.984 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 04

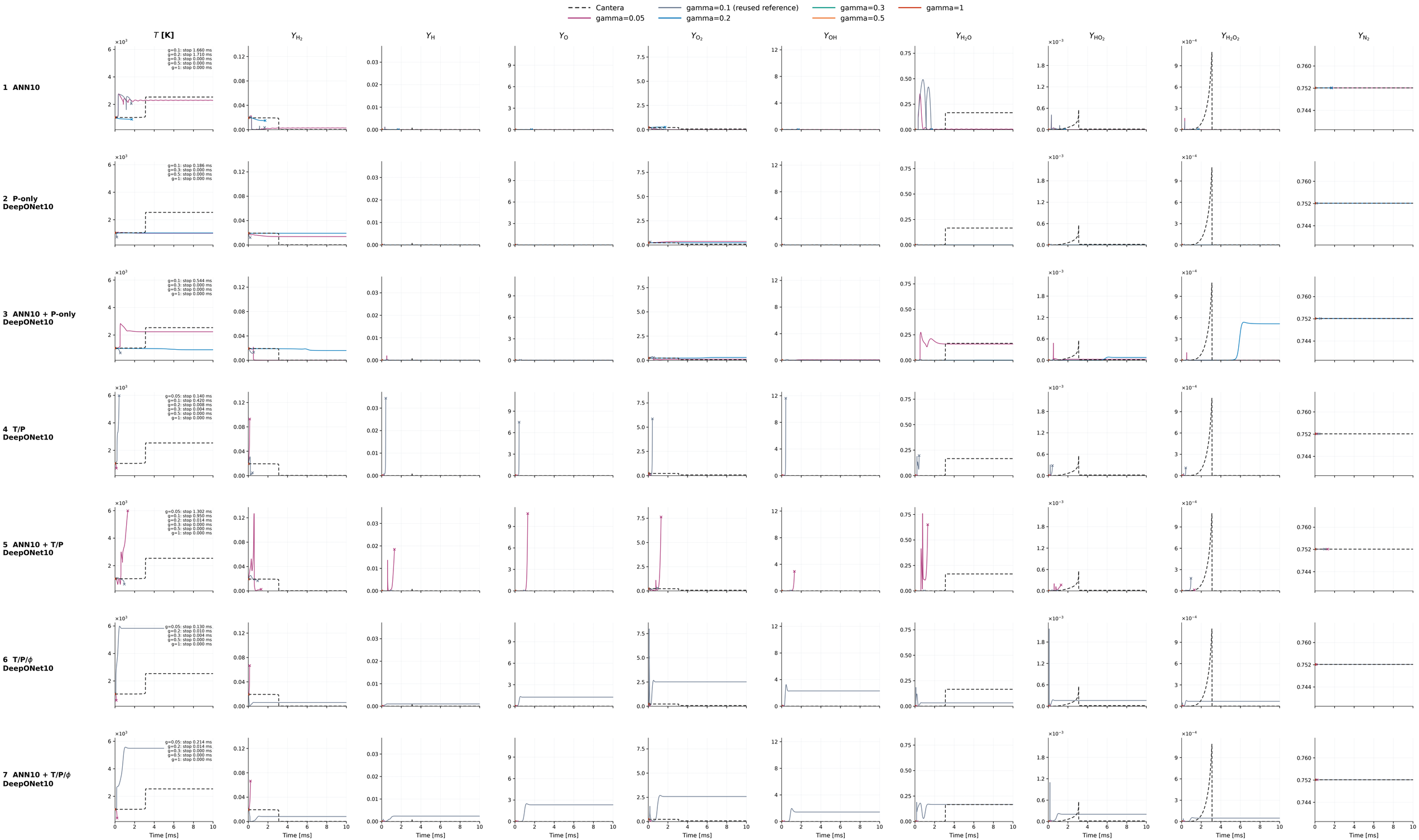
Raw | Cantera vs NN NRMSE | T0=1317.5K, P0=1.244atm, phi0=1.984

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 9.404 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.440 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.814 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 1.168 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.922 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 4.464 ms
2 P-only DeepONet10	0.2	0.0676	0.326	0.259	0.479	0.142	0.44	0.0809	0.409	2.24	3.06e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 1.280 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.240 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 4.964 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 3.810 ms
3 ANN10 + P-only DeepONet10	0.3	0.0135	0.0785	0.114	0.2	0.0814	0.204	0.0225	0.231	2.68	6.13e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	0.0293	0.0962	1.82	3.7	1.87	0.518	0.0541	5.71	0.395	3.22e-15	Complete
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.772 ms
4 T/P DeepONet10	0.2	0.02	0.0165	0.206	0.336	0.176	0.115	0.0227	0.424	0.229	3.06e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.0443	0.0251	0.419	1.06	0.275	0.545	0.0348	0.724	0.486	3.06e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.556 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.344 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.81	0.784	3.76	21.7	14.6	28.9	0.43	5.38	6.44	3.22e-15	Complete
6 T/P/ϕ DeepONet10	0.1	0.0309	0.0188	0.215	0.559	0.164	0.406	0.0425	0.406	0.363	6.13e-16	Complete
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.596	0.746	6.44	37.4	20.2	31.3	0.949	2.36	8.11	3.06e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0296	0.0176	0.223	0.634	0.106	0.427	0.047	0.294	0.389	1.53e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	4.36e-03	0.0106	0.105	0.182	0.0675	0.097	0.0168	0.192	0.126	3.06e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 05

T_0=1049.0 K P_0=4.666 atm ϕ_0=0.674 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 05

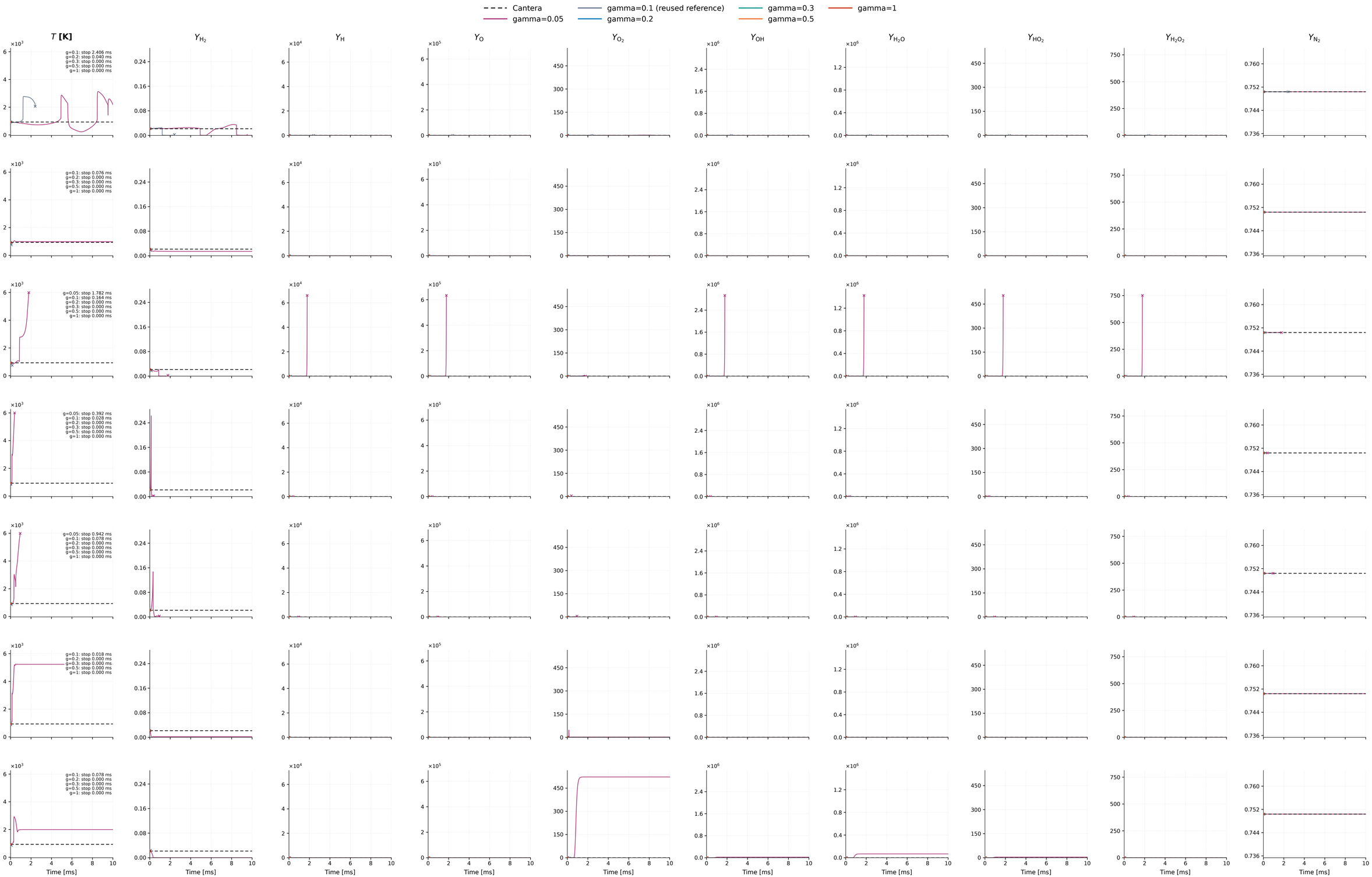
Raw | Cantera vs NN NRMSE | T0=1049.0K, P0=4.666atm, phi0=0.674

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.323	0.9	1.32	1.25	0.9	2.16	1.03	1.02	1	2.95e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.660 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.710 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.577	1.07	1	1	1.73	1	1	0.998	1	2.95e-15	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
2 P-only DeepONet10	0.2	0.569	1.46	1	1	0.917	1	1	0.995	1	2.95e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.324	0.904	1.53	1.07	0.527	8.39	0.676	0.873	0.997	2.80e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.544 ms
3 ANN10 + P-only DeepONet10	0.2	0.598	1.31	1	1	1.17	1	0.996	1.18	2.29	2.95e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.140 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.420 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.302 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.950 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.014 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.130 ms
6 T/P/ϕ DeepONet10	0.1	1.71	0.835	29.8	1.22e+03	16.9	336	0.817	2.37	0.961	1.48e-16	Complete
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.214 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	1.5	0.903	68.8	2.19e+03	16.8	203	0.602	2.85	0.951	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.014 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 07

T_0=930.6 K P_0=3.722 atm ϕ_0=0.754 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 07

Raw | Cantera vs NN NRMSE | T0=930.6K, P0=3.722atm, phi0=0.754

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.927	0.522	1.61e+08	1.58e+09	2.64	2.46e+10	8.20e+06	303	569	7.40e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.406 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.040 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.06	0.329	0.988	42.4	0.852	64.3	0.946	0.861	0.994	7.40e-16	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.782 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.164 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.392 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.942 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	4.56	0.883	3.31e+08	1.73e+11	9.55	1.17e+11	3.58e+06	349	56.1	7.40e-16	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	1.15	0.986	1.02e+11	2.34e+11	2.18e+03	4.52e+15	3.84e+11	7.94e+06	7.45e+03	2.96e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 18

T_0=1069.1 K P_0=2.170 atm ϕ_0=1.544 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 18

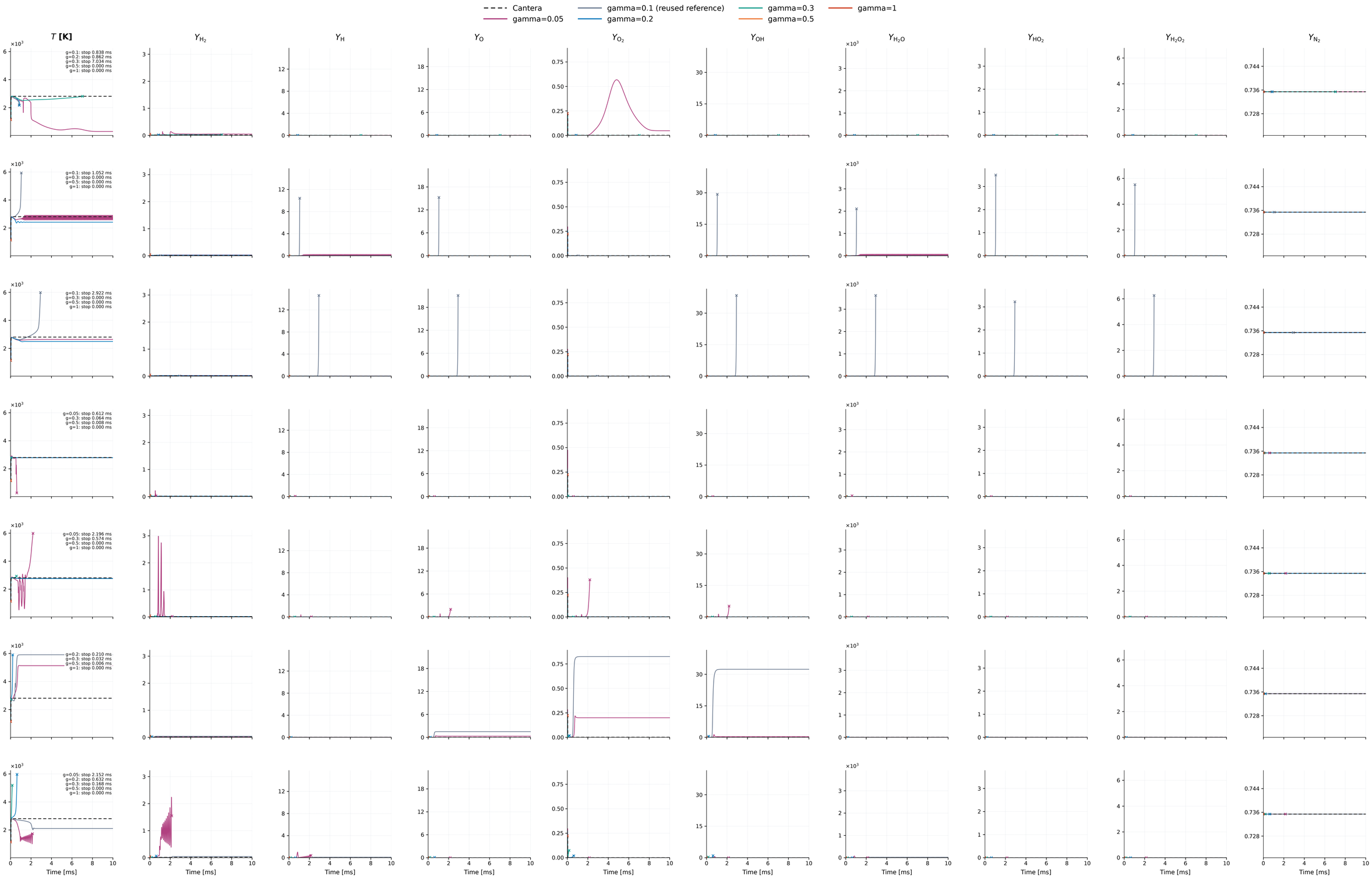
Raw | Cantera vs NN NRMSE | T0=1069.1K, P0=2.170atm, phi0=1.544

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.393	3.88	0.846	0.886	1.68	5.09	0.848	4.12	68.7	1.21e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.382 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.178 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.892 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.124 ms
2 P-only DeepONet10	0.2	0.0848	0.299	0.425	0.947	0.328	0.549	0.0938	0.919	1.06	7.56e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.724 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 4.796 ms
3 ANN10 + P-only DeepONet10	0.2	0.0794	0.342	0.549	0.944	0.346	0.618	0.0957	1.05	1.2	3.03e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.554 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.0684	0.26	1.41	0.924	0.583	2.11	1.35	1.24	1.51	1.66e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.198 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.072 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0729	54	6.15e+03	4.6e+03	0.419	292	214	1.31	1.85	1.66e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.144 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.050 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 32

T_0=1140.7 K P_0=3.354 atm ϕ_0=1.463 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 32

Raw | Cantera vs NN NRMSE | T0=1140.7K, P0=3.354atm, phi0=1.463

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.744	2.86	0.961	0.966	17.4	0.949	0.945	1.81	61.7	9.06e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.838 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.862 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 7.034 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.0507	1.09	129	0.955	0.263	2.52	138	18.6	739	9.06e-16	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.052 ms
2 P-only DeepONet10	0.2	0.137	0.651	0.134	0.87	0.275	0.709	0.483	0.699	6.1	0	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.0607	0.815	16.7	0.869	0.303	0.122	20.4	0.805	54	1.06e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.922 ms
3 ANN10 + P-only DeepONet10	0.2	0.109	0.415	0.483	0.85	0.163	0.692	0.233	0.334	1.34	4.53e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.612 ms
4 T/P DeepONet10	0.1	0.0202	0.0859	0.158	0.382	0.588	0.0956	0.0328	1.22	0.997	1.06e-15	Complete
4 T/P DeepONet10	0.2	0.0253	0.107	0.15	0.394	0.727	0.0961	0.0382	1.05	0.935	0	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.064 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.196 ms
5 ANN10 + T/P DeepONet10	0.1	0.0163	0.0579	0.204	0.378	0.395	0.172	0.0257	1.12	1.1	1.06e-15	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0275	0.092	0.24	0.465	0.646	0.196	0.0351	1.18	0.999	4.53e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.574 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.801	0.171	6.33	456	14	41.6	0.922	2.92	4.94	9.06e-16	Complete
6 T/P/ϕ DeepONet10	0.1	1.07	1.94	2.39	2.27e+03	58	3.86e+03	0.953	23.8	421	1.06e-15	Complete
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.032 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.006 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.152 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.224	2.05	102	0.727	0.326	0.829	75.3	1.85	30.6	1.06e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.632 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.168 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 39

T_0=1250.1 K P_0=2.297 atm ϕ_0=0.646 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 39

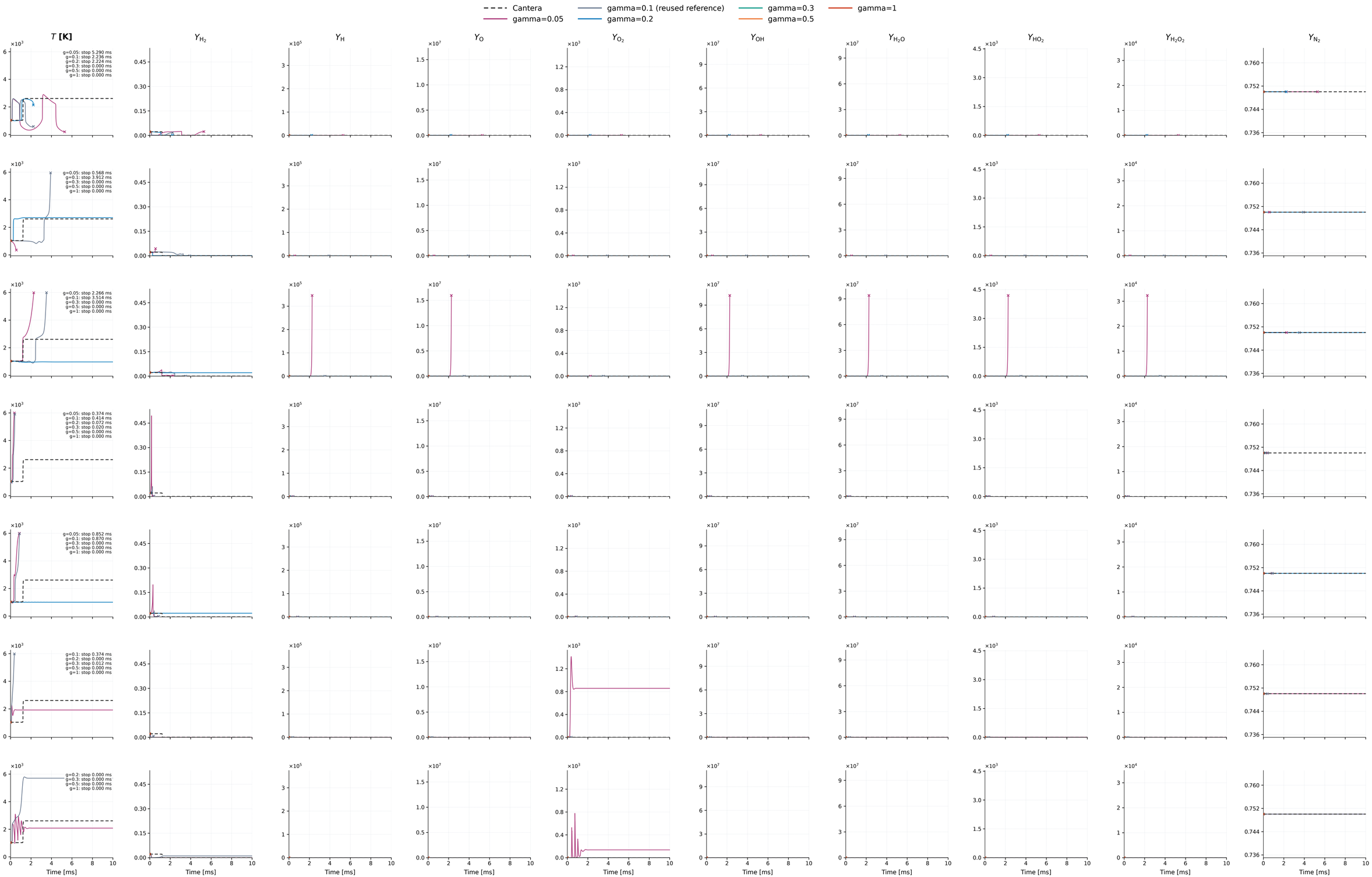
Raw | Cantera vs NN NRMSE | T0=1250.1K, P0=2.297atm, phi0=0.646

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 4.330 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.082 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.074 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.404 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.348	0.506	3.77	6.26e+03	1.48e+04	8.02e+04	1.11e+03	1.91e+05	9.29e+04	3.24e-15	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.704 ms
2 P-only DeepONet10	0.2	0.052	0.0715	0.404	0.885	0.344	0.447	0.206	0.22	0.578	1.47e-16	Complete
2 P-only DeepONet10	0.3	0.0204	0.139	0.328	0.202	0.0652	0.14	0.0148	0.138	0.104	0	Complete
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.156	0.43	0.764	16	52.4	53.2	4.75	157	68.1	3.39e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.534 ms
3 ANN10 + P-only DeepONet10	0.2	0.0463	0.176	0.222	0.848	0.559	0.517	0.286	0.448	0.783	1.03e-15	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0289	0.121	0.333	0.307	0.0179	0.213	0.0136	0.144	0.079	1.03e-15	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.402 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.228 ms
4 T/P DeepONet10	0.2	0.0484	0.45	0.881	0.699	0.0344	0.357	0.022	0.308	0.236	1.47e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.246 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.874 ms
5 ANN10 + T/P DeepONet10	0.1	0.225	0.427	126	12	214	167	67.7	97.8	29.2	5.90e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0242	0.264	0.4	0.258	0.0839	0.114	0.0477	0.128	0.0999	1.03e-15	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.540 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.264	0.525	1.22e+04	2.37e+04	3.99e+04	5.84e+06	4.89e+05	5.80e+05	2.13e+03	3.24e-15	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.334 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.264 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.176	0.506	7.04e+03	8.26e+03	6.74e+03	1.78e+06	2.26e+05	1.11e+05	2.63e+03	3.39e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.452 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.532 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.234 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 44

T_0=1031.1 K P_0=2.272 atm ϕ_0=0.770 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 44

Raw | Cantera vs NN NRMSE | T0=1031.1K, P0=2.272atm, phi0=0.770

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 5.290 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.236 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 2.224 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.568 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.912 ms
2 P-only DeepONet10	0.2	0.201	0.842	0.881	0.675	0.606	0.457	0.34	0.954	1	0	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.266 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.514 ms
3 ANN10 + P-only DeepONet10	0.2	0.622	2.49	1	1	1.84	1	1	0.913	0.914	0	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.374 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.414 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.072 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.852 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.870 ms
5 ANN10 + T/P DeepONet10	0.2	0.608	2.68	1	1	1.72	1	1	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.287	0.963	4.97e+03	8.79e+03	9.21e+03	4.31e+06	2.17e+05	2.05e+05	203	4.44e-16	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.374 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.259	0.93	1.76e+03	2.78e+03	1.52e+03	1.00e+06	6.12e+04	3.36e+04	142	2.96e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	1.21	1.51	31.7	859	15.7	160	0.333	6.13	5.11	2.22e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 45

T_0=1181.9 K P_0=4.842 atm ϕ_0=1.099 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 45

Raw | Cantera vs NN NRMSE | T0=1181.9K, P0=4.842atm, phi0=1.099

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.168	1.04	0.971	0.984	0.622	0.708	0.778	0.86	0.959	2.69e-15	Complete
1 ANN10	0.1	0.154	1.63	0.954	0.964	0.621	0.93	0.638	0.705	0.788	1.49e-16	Complete
1 ANN10	0.2	0.208	0.443	0.971	0.705	0.379	0.892	0.312	0.737	0.908	0	Complete
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 3.326 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.466 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
2 P-only DeepONet10	0.2	0.0112	0.119	0.142	0.212	0.331	0.0598	0.0522	0.81	0.739	0	Complete
2 P-only DeepONet10	0.3	0.0322	0.126	0.311	0.315	0.341	0.2	0.0787	0.764	0.646	0	Complete
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.0508	1.86	42.9	0.914	0.69	0.932	14.3	1.55	20.5	2.69e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.850 ms
3 ANN10 + P-only DeepONet10	0.2	0.0222	0.146	0.242	0.184	0.299	0.157	0.0204	0.811	0.821	4.48e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	0.03	0.13	0.317	0.301	0.309	0.209	0.0393	0.765	0.635	2.99e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.494 ms
4 T/P DeepONet10	0.1	0.0217	0.262	0.261	0.199	0.511	0.0576	0.0293	0.84	0.605	1.49e-16	Complete
4 T/P DeepONet10	0.2	0.719	37.2	2.78e+13	1.38e+14	6.67e+10	6.54e+12	5.40e+11	1.37e+10	2.20e+08	0	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.806 ms
5 ANN10 + T/P DeepONet10	0.1	0.0214	0.315	0.312	0.16	0.505	0.116	0.0315	0.885	0.653	1.49e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0182	0.219	0.154	0.15	0.535	0.0488	0.0516	0.784	0.6	4.48e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.296 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.346	0.944	71.7	123	2.97e+03	2.77e+05	1.13e+04	1.96e+04	2.42e+03	2.69e-15	Complete
6 T/P/ϕ DeepONet10	0.1	0.897	2.9	2.57	818	1.26e+03	788	5.08	31.5	155	1.49e-16	Complete
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.098 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.048 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.733	0.577	10.8	746	71.5	6.88	4.07	2.19	1.06	2.69e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.805	3.6	3.74	1.03e+03	80.1	212	0.391	12.5	67.8	1.49e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.206 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.152 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 53

T_0=1257.8 K P_0=2.103 atm ϕ_0=1.799 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 53

Raw | Cantera vs NN NRMSE | T0=1257.8K, P0=2.103atm, phi0=1.799

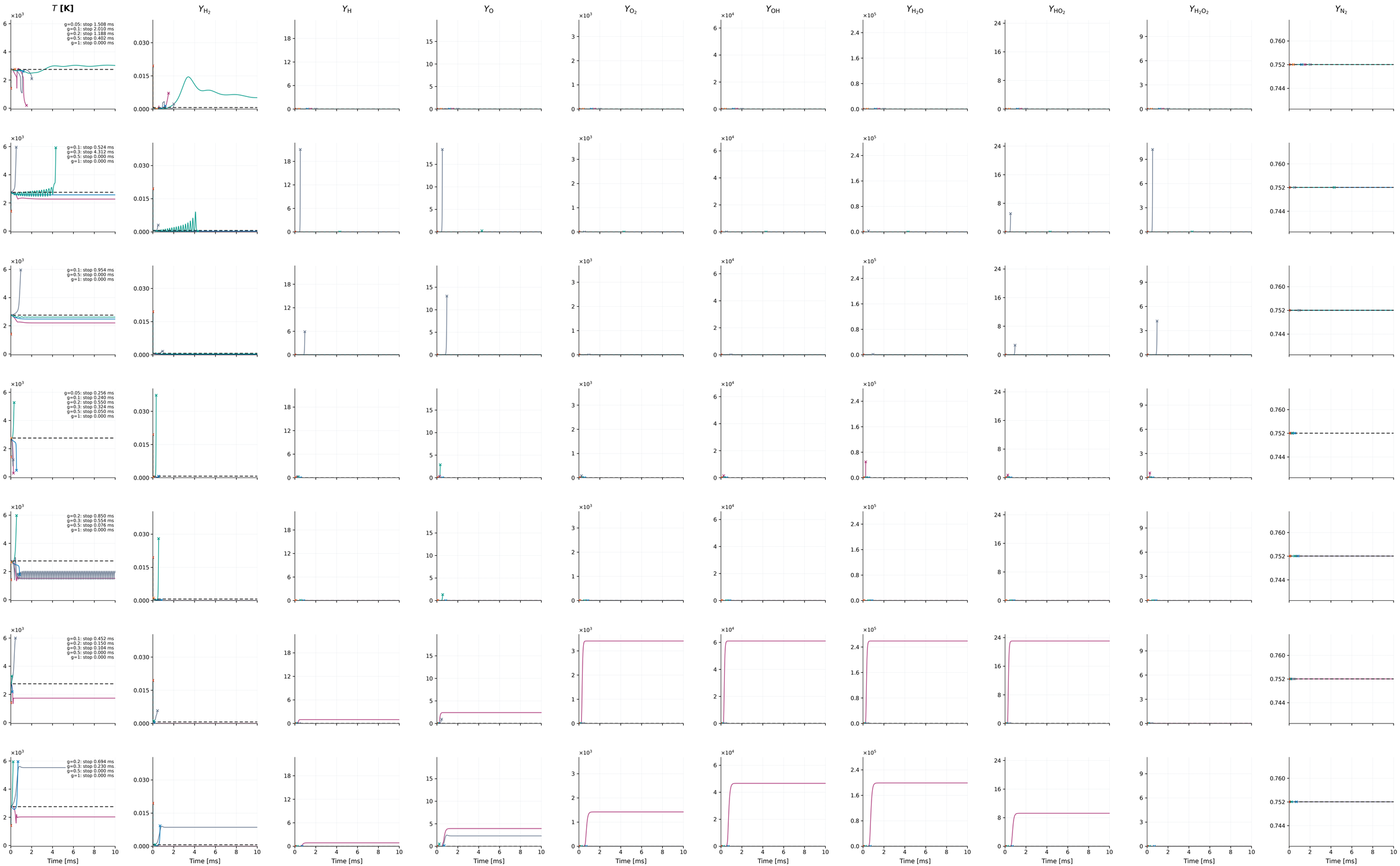
Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.376	8.53	0.758	0.732	0.828	3.79	0.931	1.42	7.18	6.10e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.320 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.290 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 3.392 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.006 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.148 ms
2 P-only DeepONet10	0.2	0.101	0.319	0.357	0.719	0.075	0.634	0.116	0.153	2.15	0	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.018 ms
3 ANN10 + P-only DeepONet10	0.1	0.0124	0.292	0.396	0.464	0.0498	0.204	0.136	0.121	0.506	7.62e-16	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0905	0.311	0.483	0.771	0.077	0.673	0.0977	0.165	0.547	7.62e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	0.0568	0.038	0.352	0.141	0.225	0.12	0.307	0.389	0.435	6.10e-16	Complete
4 T/P DeepONet10	0.1	0.0377	0.0241	0.286	0.477	0.312	0.332	0.0163	0.592	0.456	7.62e-16	Complete
4 T/P DeepONet10	0.2	0.0249	0.0568	0.163	0.345	0.452	0.0437	0.112	0.856	0.582	0	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.256 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.0532	0.108	0.353	0.141	0.163	0.333	0.489	0.335	0.334	1.07e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0356	0.0124	0.425	0.598	0.201	0.426	0.106	0.415	0.373	7.62e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0213	0.0267	0.149	0.3	0.333	0.18	0.0343	0.693	0.525	7.62e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.520 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.252 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.294 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.272 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.072 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.770 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.05	0.249	0.583	0.893	0.258	0.86	0.241	0.547	1.1	7.62e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.752 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 57

T_0=1412.0 K P_0=4.083 atm ϕ_0=0.677 | Raw self-fed rollout, $\Delta t = 2\,\mu$s





New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 57

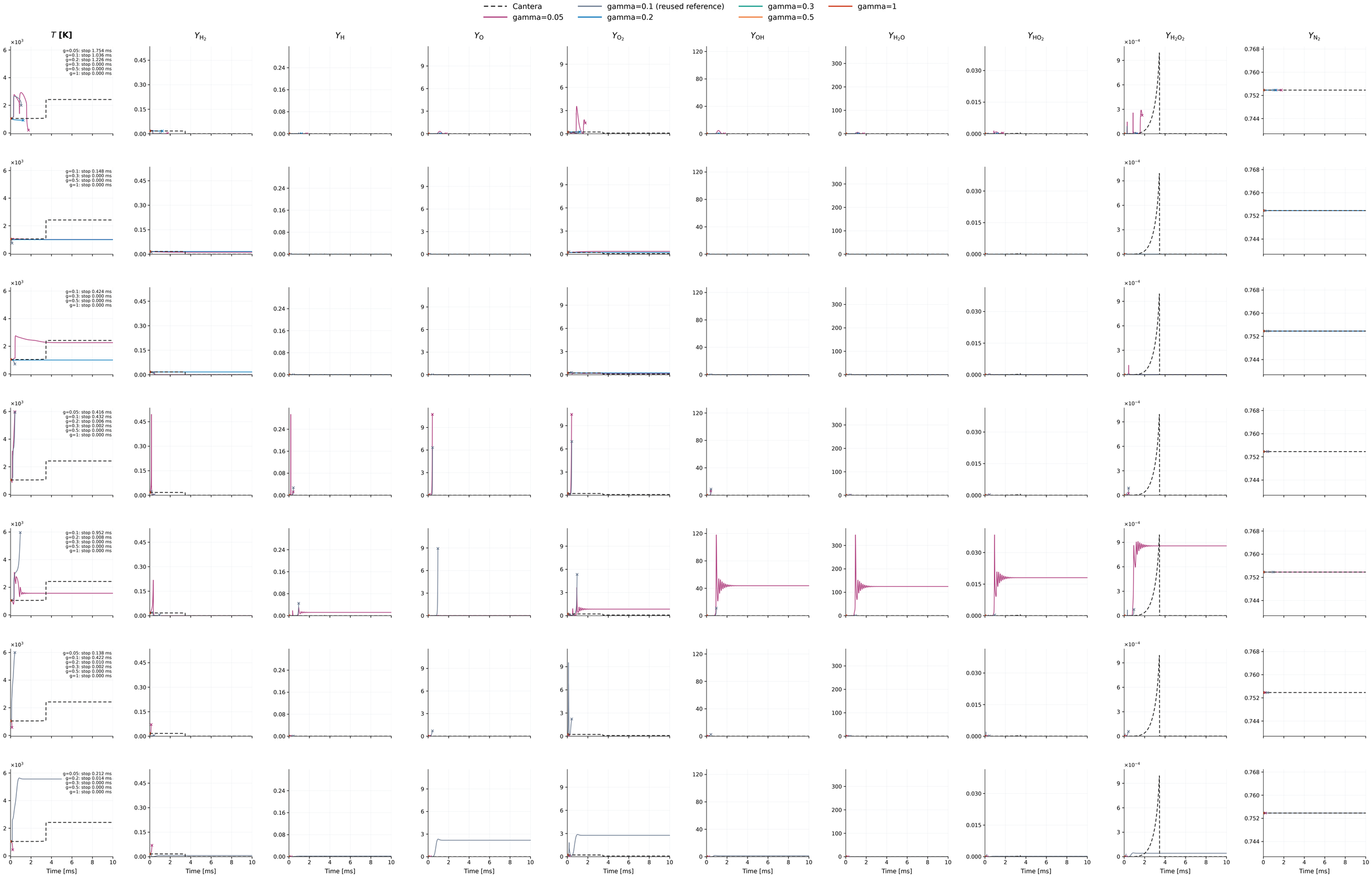
Raw | Cantera vs NN NRMSE | T0=1412.0K, P0=4.083atm, phi0=0.677

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.508 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.010 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.188 ms
1 ANN10	0.3	0.0821	8.13	1.65	0.451	0.899	0.894	5.95	0.798	0.849	0	Complete
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.402 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.168	0.76	0.931	0.694	1.57	5.86	0.817	0.688	0.49	0	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.524 ms
2 P-only DeepONet10	0.2	0.0646	0.38	0.626	0.561	0.0497	0.4	0.0651	0.219	0.0454	4.43e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 4.312 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.193	0.728	0.94	0.873	0.644	4.14	0.0651	0.342	0.275	4.43e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.954 ms
3 ANN10 + P-only DeepONet10	0.2	0.0963	0.494	0.761	0.665	0.0265	0.505	0.0296	0.409	0.286	8.86e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	0.05	0.306	0.497	0.483	0.0935	0.314	0.0465	0.3	0.234	1.03e-15	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.256 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.240 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.550 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.324 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.050 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.439	0.804	71	1.39	7.11	1.84e+03	524	417	403	4.43e-16	Complete
5 ANN10 + T/P DeepONet10	0.1	0.332	0.813	56.4	0.932	63.3	136	59.5	57.4	32.6	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.850 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.554 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.367	0.823	7.43e+03	708	4.78e+04	3.98e+06	1.60e+06	8.40e+05	2.13e+03	0	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.452 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.104 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.259	0.814	6.55e+03	1.13e+03	1.95e+04	2.97e+06	1.20e+06	3.29e+05	2.14e+03	4.43e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.977	9.31	18.2	665	32.5	95.2	0.146	6.37	26.8	1.48e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.694 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.230 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 64

T_0=1045.3 K P_0=4.301 atm ϕ_0=0.595 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 64

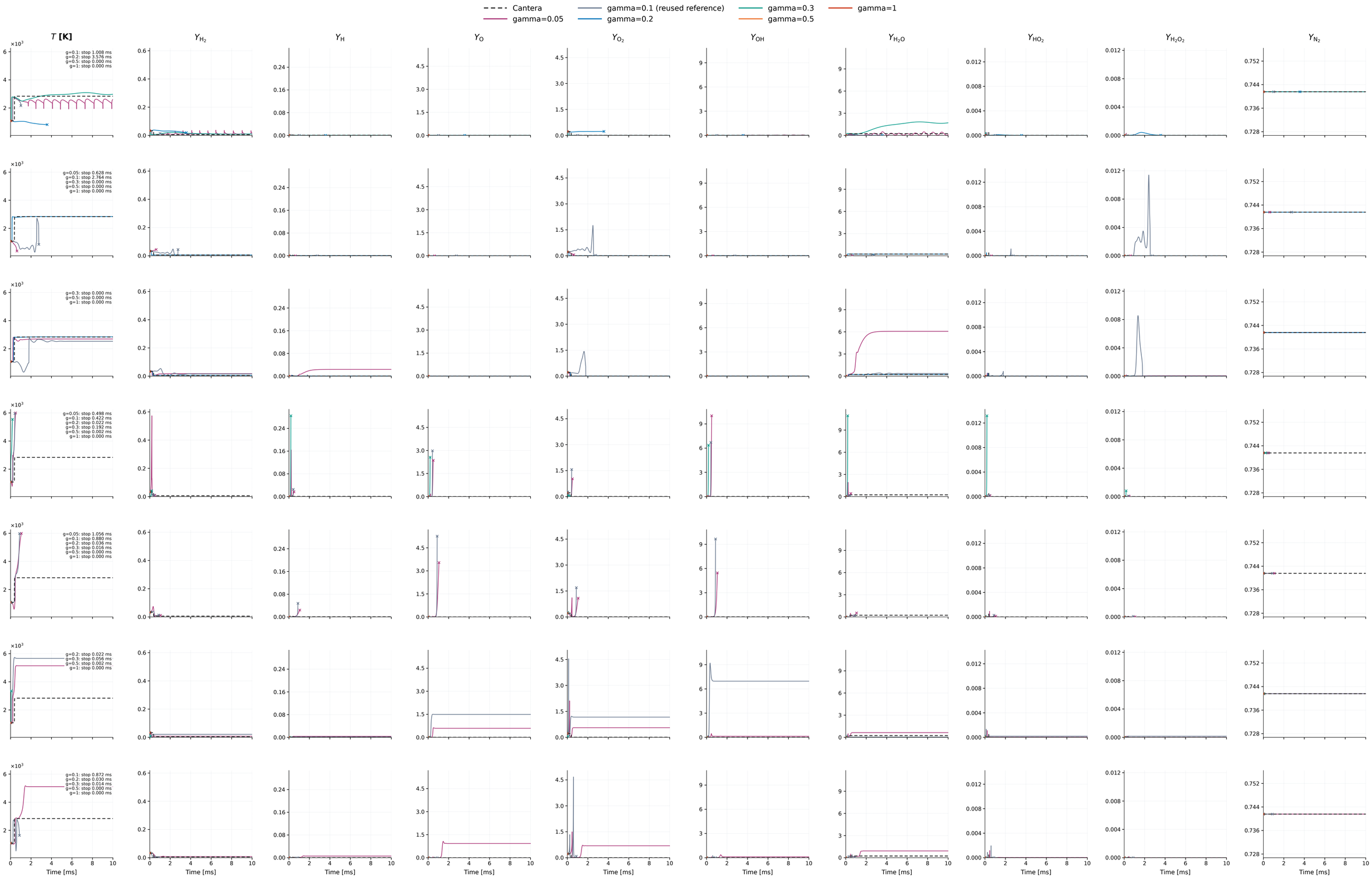
Raw | Cantera vs NN NRMSE | T0=1045.3K, P0=4.301atm, phi0=0.595

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.754 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.036 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.226 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	0.556	1	1	1	1.55	1	1	0.998	1	1.47e-16	Complete
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
2 P-only DeepONet10	0.2	0.56	1.36	1	1	0.751	1	1	0.999	1	1.47e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.388	0.929	1.43	2.42	0.469	17.7	1.05	0.935	0.997	2.95e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.424 ms
3 ANN10 + P-only DeepONet10	0.2	0.558	1.36	1	1	0.747	1	1	0.996	1	1.03e-15	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.416 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.432 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.006 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.396	1.55	640	16	4.68	9.08e+03	996	295	5.21	2.95e-16	Complete
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.952 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.138 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.422 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.212 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	1.72	0.836	120	3.17e+03	16.6	271	0.621	3.01	0.949	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.014 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 68

T_0=1086.3 K P_0=3.702 atm ϕ_0=1.168 | Raw self-fed rollout, $\Delta t = 2\,\mu$s



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 68

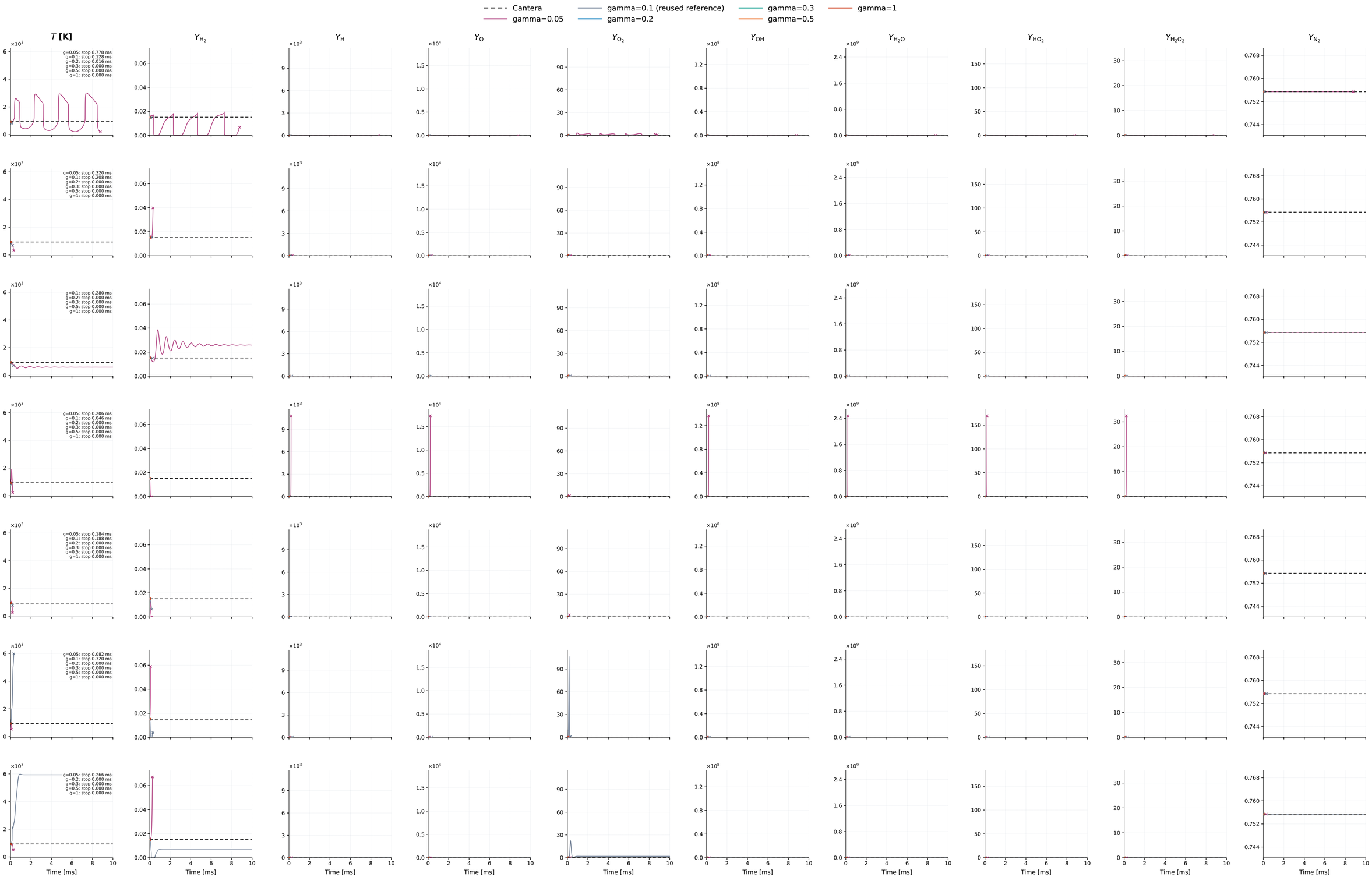
Raw | Cantera vs NN NRMSE | T0=1086.3K, P0=3.702atm, phi0=1.168

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.145	0.88	0.954	0.996	0.661	1.15	0.653	1.4	1.68	2.99e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.008 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 3.576 ms
1 ANN10	0.3	0.12	1.06	0.432	1.45	0.865	0.987	5.07	1.21	1.04	0	Complete
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.628 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.764 ms
2 P-only DeepONet10	0.2	0.0923	0.485	0.228	0.3	0.752	0.169	0.162	1.07	0.999	1.50e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.0901	1.42	38.8	0.938	0.629	0.37	23.4	1.2	5.15	2.99e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	0.295	1.67	0.514	0.978	5.87	0.866	0.624	2.66	163	4.49e-16	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0546	0.288	0.21	0.285	0.451	0.118	0.107	1.05	1	2.99e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.498 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.422 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.022 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.192 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.056 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.880 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.812	0.477	5.41	421	12.4	10.2	1.72	1.73	1.09	2.99e-15	Complete
6 T/P/ϕ DeepONet10	0.1	1.01	1.81	2.29	1.08e+03	26.3	575	0.851	7.61	18.2	4.49e-16	Complete
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.022 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.056 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.761	0.329	9.08	632	14.6	8.43	2.77	1.94	1.12	2.99e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.872 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.030 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.014 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 69

T_0=935.0 K P_0=2.045 atm ϕ_0=0.523 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 69

Raw | Cantera vs NN NRMSE | T0=935.0K, P0=2.045atm, phi0=0.523

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 8.778 ms
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.128 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.320 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.36	0.72	1	1	0.312	1	1	1	1	1.47e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.206 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.184 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.188 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.082 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.320 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.266 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	5.19	0.582	9.65e+08	1.69e+11	12.3	4.23e+11	2.17e+06	829	349	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 73

T_0=1106.0 K P_0=1.911 atm ϕ_0=1.915 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 73

Raw | Cantera vs NN NRMSE | T0=1106.0K, P0=1.911atm, phi0=1.915

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.384	11.5	2.63	3.39	0.314	15.6	2.39	0.498	2.09	2.90e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.918 ms
1 ANN10	0.2	0.0466	0.221	0.584	0.736	0.33	0.294	0.274	0.783	0.889	3.06e-16	Complete
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.964 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 3.434 ms
2 P-only DeepONet10	0.2	0.0491	0.271	0.241	0.572	0.199	0.373	0.0965	0.487	0.932	3.06e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 2.370 ms
3 ANN10 + P-only DeepONet10	0.1	0.06	0.5	0.694	2.01	0.0571	0.946	0.199	0.207	0.307	7.64e-16	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0479	0.267	0.27	0.692	0.278	0.435	0.0955	0.673	0.867	1.22e-15	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.334 ms
4 T/P DeepONet10	0.1	0.0213	0.271	0.332	0.788	0.495	0.316	0.107	1.23	1.2	7.64e-16	Complete
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.390 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.112 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	8.58e-03	0.0735	0.892	1.08	0.273	0.903	0.271	0.575	0.679	2.90e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0125	0.0563	0.39	0.725	0.384	0.262	0.213	0.953	1.05	7.64e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.434 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.266 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.312 ms
6 T/P/ϕ DeepONet10	0.2	0.182	1.11	19.6	21.7	0.594	0.408	5.49	1.35	1.18	3.06e-16	Complete
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.779	7.24	5.39e+03	1.44e+03	2.99	178	14.9	1.7	4.6	2.90e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0314	0.167	0.638	1.09	0.399	0.587	0.172	0.993	1.07	7.64e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.734 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 74

T_0=1222.9 K P_0=4.971 atm ϕ_0=1.766 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 74

Raw | Cantera vs NN NRMSE | T0=1222.9K, P0=4.971atm, phi0=1.766

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.113	0.567	0.974	0.934	0.245	0.685	1.71	0.565	0.879	1.67e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.666 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 2.708 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 4.606 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.502 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.342 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.216 ms
2 P-only DeepONet10	0.3	0.0294	0.0284	0.252	0.505	0.275	0.282	0.0444	0.637	0.888	0	Complete
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.706 ms
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.354 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.336 ms
3 ANN10 + P-only DeepONet10	0.3	0.0377	0.0428	0.359	0.606	0.298	0.385	0.051	0.679	0.86	9.13e-16	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	0.0658	0.044	0.436	1.41	0.569	0.507	0.106	1.03	0.991	1.67e-15	Complete
4 T/P DeepONet10	0.1	0.0146	0.0312	0.19	0.55	0.598	0.207	0.0463	1.03	0.965	1.83e-15	Complete
4 T/P DeepONet10	0.2	0.0393	0.0402	0.362	0.708	0.69	0.421	0.0339	0.991	0.957	7.61e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	0.294	117	649	1.63e+03	0.656	450	18.1	1.23	2.63	1.67e-15	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0414	0.092	0.578	0.866	0.539	0.599	0.208	1.07	0.987	1.37e-15	Complete
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.480 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.314 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.292 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.044 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.408 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.678 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.606 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 76

T_0=1523.6 K P_0=3.626 atm ϕ_0=1.349 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 76

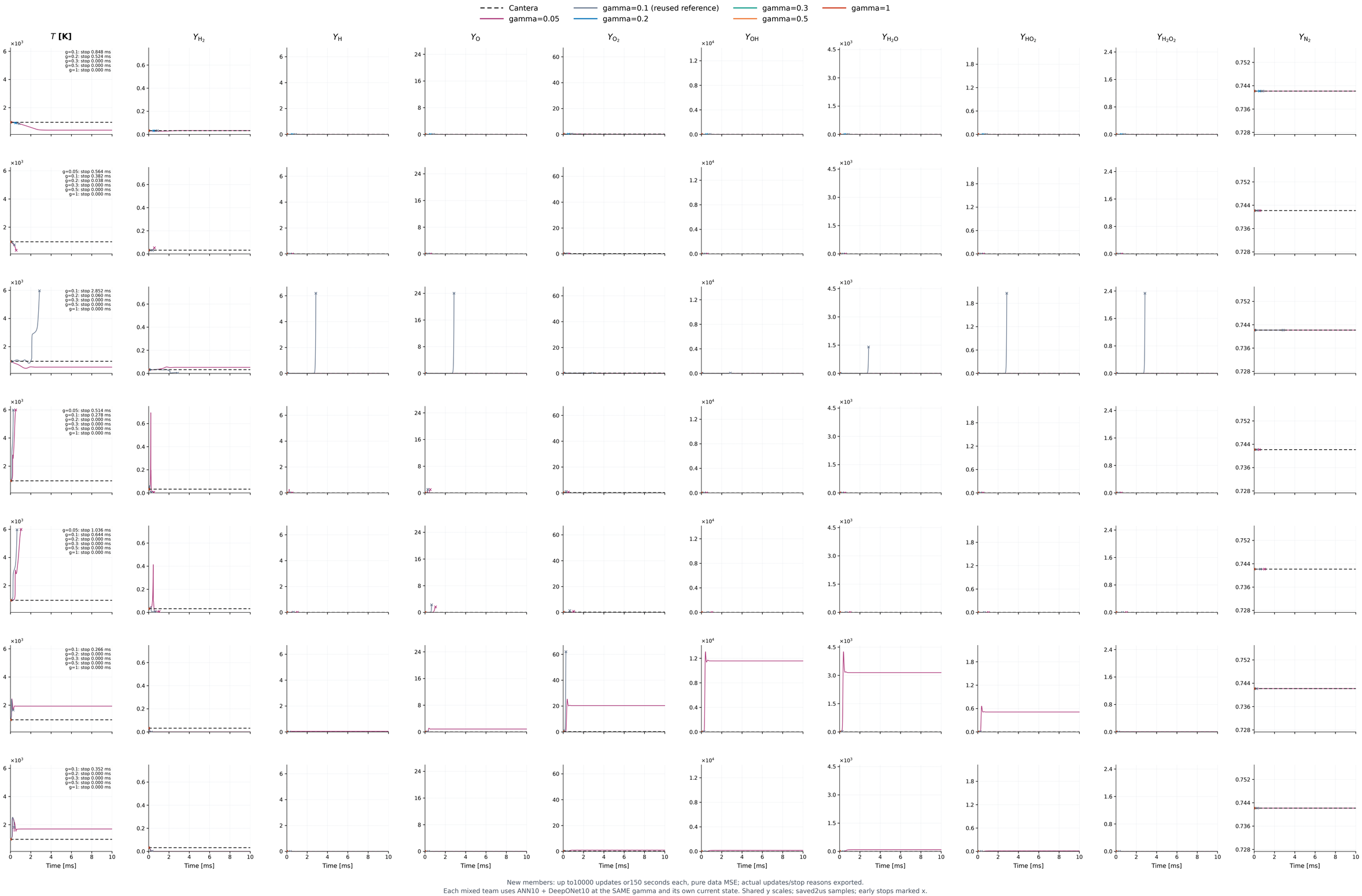
Raw | Cantera vs NN NRMSE | T0=1523.6K, P0=3.626atm, phi0=1.349

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.157	1.09	0.944	0.966	0.665	0.645	1.17	0.936	0.954	2.56e-15	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.202 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.582 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.500 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.572 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.964 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.718 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.994 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.536 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.109	0.909	11.3	0.939	0.713	0.59	20.2	0.44	46.4	2.11e-15	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 4.378 ms
3 ANN10 + P-only DeepONet10	0.2	0.0492	0.0552	0.358	0.593	0.444	0.418	0.0626	0.661	0.784	7.52e-16	Complete
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.968 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.600 ms
4 T/P DeepONet10	0.1	0.0553	0.0153	0.391	0.634	0.466	0.448	0.0778	0.653	0.534	6.02e-16	Complete
4 T/P DeepONet10	0.2	0.0484	0.0624	0.401	0.588	0.367	0.376	0.0193	0.566	0.371	7.52e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.316 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.984 ms
5 ANN10 + T/P DeepONet10	0.1	0.0546	8.51e-03	0.396	0.647	0.485	0.475	0.0889	0.671	0.58	6.02e-16	Complete
5 ANN10 + T/P DeepONet10	0.2	0.061	0.0527	0.472	0.678	0.461	0.469	0.0584	0.647	0.506	7.52e-16	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.154 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.030 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.558 ms
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.612 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.094 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.554 ms
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.714	2.11	1.21	590	89.7	351	0.664	27.1	163	6.02e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.274 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 78

T_0=962.3 K P_0=2.443 atm ϕ_0=1.140 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 78

Raw | Cantera vs NN NRMSE | T0=962.3K, P0=2.443atm, phi0=1.140

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.517	0.0566	1	1	0.478	1	1	1	1	8.97e-16	Complete
1 ANN10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.848 ms
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.524 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.564 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.382 ms
2 P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.416	0.577	1	1	0.607	1	1	1	1	8.97e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 2.852 ms
3 ANN10 + P-only DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
3 ANN10 + P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.514 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.278 ms
4 T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.036 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.644 ms
5 ANN10 + T/P DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	1.01	0.995	2.44e+08	3.00e+09	88.1	7.88e+13	8.29e+08	1.07e+05	937	8.97e-16	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.266 ms
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.778	0.989	3.03e+06	1.36e+08	3.37	1.11e+12	2.41e+07	2.7e+03	107	8.97e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep: Cantera vs NN | Test IC 88

T_0=1502.1 K P_0=4.980 atm ϕ_0=0.859 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera
gamma=0.05
gamma=0.1 (reused reference)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

New members: up to10000 updates or150 seconds each, pure data MSE; actual updates/stop reasons exported.
Each mixed team uses ANN10 + DeepONet10 at the SAME gamma and its own current state. Shared y scales; saved2us samples; early stops marked x.
Gamma=.1 models are frozen controls; historical training time is separate. Raw/repaired retain the original decoding and physical correction.

Box-Cox gamma sweep | Test IC 88

Raw | Cantera vs NN NRMSE | T0=1502.1K, P0=4.980atm, phi0=0.859

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.195	1.55	0.966	0.978	0.946	0.56	0.963	0.969	1.01	1.48e-16	Complete
1 ANN10	0.1	0.165	2.68	0.929	0.95	0.951	0.9	0.71	0.956	0.948	8.90e-16	Complete
1 ANN10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 1.382 ms
1 ANN10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 2.732 ms
1 ANN10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.154 ms
1 ANN10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.534 ms
2 P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.418 ms
2 P-only DeepONet10	0.2	0.0552	0.409	0.562	0.47	0.148	0.349	0.104	0.396	0.478	2.97e-16	Complete
2 P-only DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 1.098 ms
2 P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	0.05	0.205	0.698	0.778	0.927	1.28	1.17	0.651	1.42	2.26	5.94e-16	Complete
3 ANN10 + P-only DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.752 ms
3 ANN10 + P-only DeepONet10	0.2	0.0641	0.455	0.628	0.523	0.154	0.404	0.104	0.44	0.502	1.48e-15	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0808	0.559	0.723	0.589	0.171	0.44	0.101	0.312	0.349	1.34e-15	Complete
3 ANN10 + P-only DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
3 ANN10 + P-only DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 0.454 ms
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.460 ms
4 T/P DeepONet10	0.2	0.0889	0.494	0.73	0.603	0.0741	0.42	0.0913	0.403	0.297	2.97e-16	Complete
4 T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.054 ms
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	0.05	—	—	—	—	—	—	—	—	—	—	Stop 1.044 ms
5 ANN10 + T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.910 ms
5 ANN10 + T/P DeepONet10	0.2	0.0718	0.419	0.627	0.558	0.164	0.398	0.126	0.409	0.327	1.48e-15	Complete
5 ANN10 + T/P DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.126 ms
5 ANN10 + T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
5 ANN10 + T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	0.05	0.702	0.538	4.66	324	47.3	3.31	2.87	0.391	0.192	1.48e-16	Complete
6 T/P/ϕ DeepONet10	0.1	0.896	3.88	2.42	341	53.6	140	0.713	5.19	42.7	8.90e-16	Complete
6 T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
6 T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.032 ms
6 T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.326	0.953	572	129	7.08e+03	7.14e+05	2.57e+05	8.33e+04	1.62e+03	5.94e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.807	5.08	5.33	500	48.7	88	0.136	6.07	29.2	8.90e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	—	—	—	—	—	—	—	—	—	—	Stop 0.138 ms
7 ANN10 + T/P/ϕ DeepONet10	0.3	—	—	—	—	—	—	—	—	—	—	Stop 0.062 ms
7 ANN10 + T/P/ϕ DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
7 ANN10 + T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

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